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Resilient Cloud Solutions

AWS Certified DevOps Engineer – Professional (DOP-C02) ｜ 官方考試權重 15%

第 1 題 （題庫 No.6 · 單選）

A company is using an Amazon Aurora cluster as the data store for its application. The Aurora cluster is configured with a single DB instance. The application performs read and write operations on the database by using the cluster ' s instance endpoint. The company has scheduled an update to be applied to the cluster during an upcoming maintenance window. The cluster must remain available with the least possible interruption during the maintenance window. What should a DevOps engineer do to meet these requirements?

A. Add a reader instance to the Aurora cluster. Update the application to use the Aurora cluster endpoint for write operations. Update the Aurora cluster ' s reader endpoint for reads.

B. Add a reader instance to the Aurora cluster. Create a custom ANY endpoint for the cluster. Update the application to use the Aurora cluster ' s custom ANY endpoint for read and write operations.

C. Turn on the Multi-AZ option on the Aurora cluster. Update the application to use the Aurora cluster endpoint for write operations. Update the Aurora cluster's reader endpoint for reads.

D. Turn on the Multi-AZ option on the Aurora cluster. Create a custom ANY endpoint for the cluster.Update the application to use the Aurora cluster ' s custom ANY endpoint for read and write operations.

我的答案：_____

第 2 題 （題庫 No.42 · 單選）

A company has deployed a critical application in two AWS Regions. The application uses an Application Load Balancer (ALB) in both Regions. The company has Amazon Route 53 alias DNS records for both ALBs. The company uses Amazon Route 53 Application Recovery Controller to ensure that the application can fail over between the two Regions. The Route 53 ARC configuration includes a routing control for both Regions. The company uses Route 53 ARC to perform quarterly disaster recovery (DR) tests. During the most recent DR test, a DevOps engineer accidentally turned off both routing controls. The company needs to ensure that at least one routing control is turned on at all times. Which solution will meet these requirements?

A. In Route 53 ARC. create a new assertion safety rule. Apply the assertion safety rule to the two routing controls. Configure the rule with the ATLEAST type with a threshold of 1.

B. In Route 53 ARC, create a new gating safety rule. Apply the assertion safety rule to the two routing controls. Configure the rule with the OR type with a threshold of 1.

C. In Route 53 ARC, create a new resource set. Configure the resource set with an AWS: Route53: HealthCheck resource type. Specify the ARNs of the two routing controls as the target resource. Create a new readiness check for the resource set.

D. In Route 53 ARC, create a new resource set. Configure the resource set with an AWS:Route53RecoveryReadiness: DNSTargetResource resource type. Add the domain names of the two Route 53 alias DNS records as the target resource. Create a new readiness check for the resource set.

我的答案：_____

第 3 題 （題庫 No.54 · 複選（Select 3））

A company uses S3 to store images and requires multi-Region DR with two-way replication and #15-minute latency. Which steps meet the requirements? (Select THREE.)

A. Enable S3 Replication Time Control (RTC) for each replication rule.

B. Create S3 Multi-Region Access Point (active/passive).

C. Call SubmitMultiRegionAccessPointRoutes during failover.

D. Enable S3 Transfer Acceleration.

E. Use Route 53 ARC routing control.

F. Use Route 53 ARC to shift traffic during failover.

我的答案：_____

第 4 題 （題庫 No.130 · 單選）

A company is hosting a static website from an Amazon S3 bucket. The website is available to customers at example.com. The company uses an Amazon Route 53 weighted routing policy with a TTL of 1 day. The company has decided to replace the existing static website with a dynamic web application. The dynamic web application uses an Application Load Balancer (ALB) in front of a fleet of Amazon EC2 instances. On the day of production launch to customers, the company creates an additional Route 53 weighted DNS record entry that points to the ALB with a weight of 255 and a TTL of 1 hour. Two days later, a DevOps engineer notices that the previous static website is displayed sometimes when customers navigate to example. com. How can the DevOps engineer ensure that the company serves only dynamic content for example.com?

A. Delete all objects, including previous versions, from the S3 bucket that contains the static website content.

B. Update the weighted DNS record entry that points to the S3 bucket. Apply a weight of 0. Specify the domain reset option to propagate changes immediately.

C. Configure webpage redirect requests on the S3 bucket with a hostname that redirects to the ALB.

D. Remove the weighted DNS record entry that points to the S3 bucket from the example.com hosted zone. Wait for DNS propagation to become complete.

我的答案：_____

第 5 題 （題庫 No.184 · 複選（Select 3））

A DevOps engineer is building an application that uses an AWS Lambda function to query an Amazon Aurora MySQL DB cluster. The Lambda function performs only read queries. Amazon EventBridge events invoke the Lambda function. As more events invoke the Lambda function each second, the database ' s latency increases and the database ' s throughput decreases. The DevOps engineer needs to improve the performance of the application. Which combination of steps will meet these requirements? (Select THREE.)

A. Use Amazon RDS Proxy to create a proxy. Connect the proxy to the Aurora cluster reader endpoint. Set a maximum connections percentage on the proxy.

B. Implement database connection pooling inside the Lambda code. Set a maximum number of connections on the database connection pool.

C. Implement the database connection opening outside the Lambda event handler code.

D. Implement the database connection opening and closing inside the Lambda event handler code.

E. Connect to the proxy endpoint from the Lambda function.

F. Connect to the Aurora cluster endpoint from the Lambda function.

我的答案：_____

第 6 題 （題庫 No.212 · 單選）

A company has an application that is using a MySQL-compatible Amazon Aurora Multi-AZ DB cluster as the database. A cross-Region read replica has been created for disaster recovery purposes. A DevOps engineer wants to automate the promotion of the replica so it becomes the primary database instance in the event of a failure. Which solution will accomplish this?

A. Configure a latency-based Amazon Route 53 CNAME with health checks so it points to both the primary and replica endpoints. Subscribe an Amazon SNS topic to Amazon RDS failure notifications from AWS CloudTrail and use that topic to invoke an AWS Lambda function that will promote the replica instance as the primary.

B. Create an Aurora custom endpoint to point to the primary database instance. Configure the application to use this endpoint. Configure AWS CloudTrail to run an AWS Lambda function to

promote the replica instance and modify the custom endpoint to point to the newly promoted instance.

C. Create an AWS Lambda function to modify the application ' s AWS CloudFormation template to promote the replica, apply the template to update the stack, and point the application to the newly promoted instance. Create an Amazon CloudWatch alarm to invoke this Lambda function after the failure event occurs.

D. Store the Aurora endpoint in AWS Systems Manager Parameter Store. Create an Amazon EventBridge event that detects the database failure and runs an AWS Lambda function to promote the replica instance and update the endpoint URL stored in AWS Systems Manager Parameter Store. Code the application to reload the endpoint from Parameter Store if a database connection fails.

我的答案：_____

第 7 題 （題庫 No.254 · 單選）

A company is migrating its web application to AWS. The application uses WebSocket connections for real- time updates and requires sticky sessions. A DevOps engineer must implement a highly available architecture for the application. The application must be accessible to users worldwide with the least possible latency. Which solution will meet these requirements with the LEAST operational overhead?

A. Deploy an Application Load Balancer (ALB). Deploy another ALB in a different AWS Region. Enable cross-zone load balancing and sticky sessions on the ALBs. Integrate the ALBs with Amazon Route 53 latency-based routing.

B. Deploy a Network Load Balancer (NLB). Deploy another NLB in a different AWS Region. Enable cross-zone load balancing and sticky sessions on the NLBs. Integrate the NLBs with Amazon Route 53 geolocation routing.

C. Deploy a Network Load Balancer (NLB) with cross-zone load balancing enabled. Configure the NLB with IP-based targets in multiple Availability Zones. Use Amazon CloudFront for global content delivery. Implement sticky sessions by using source IP address preservation on the NLB.

D. Deploy an Application Load Balancer (ALB) for HTTP traffic. Deploy a Network Load Balancer (NLB) in each of the company ' s AWS Regions for WebSocket connections. Enable sticky sessions on the ALB. Configure the ALB to forward requests to the NLB.

我的答案：_____

第 8 題 （題庫 No.296 · 單選）

A company is running an application on Amazon EC2 instances in an Auto Scaling group. Recently an issue occurred that prevented EC2 instances from launching successfully and it took several hours for the support team to discover the issue. The support team wants to be notified by email whenever an EC2 instance does not start successfully. Which action will accomplish this?

A. Add a health check to the Auto Scaling group to invoke an AWS Lambda function whenever an instance status is impaired.

B. Configure the Auto Scaling group to send a notification to an Amazon SNS topic whenever a failed instance launch occurs.

C. Create an Amazon CloudWatch alarm that invokes an AWS Lambda function when a failed Attachinstances Auto Scaling API call is made.

D. Create a status check alarm on Amazon EC2 to send a notification to an Amazon SNS topic whenever a status check fail occurs.

我的答案：_____

第 9 題 （題庫 No.302 · 複選（Select 2））

A company recently deployed its web application on AWS. The company is preparing for a large-scale sales event and must ensure that the web application can scale to meet the demand The application ' s frontend infrastructure includes an Amazon CloudFront distribution that has an Amazon S3 bucket as an origin. The backend infrastructure includes an Amazon API Gateway API. several AWS Lambda functions, and an Amazon Aurora DB cluster The company ' s DevOps engineer conducts a load test

and identifies that the Lambda functions can fulfill the peak number of requests However, the DevOps engineer notices request latency during the initial burst of requests Most of the requests to the Lambda functions produce queries to the database A large portion of the invocation time is used to establish database connections Which combination of steps will provide the application with the required scalability? (Select TWO)

A. Configure a higher reserved concurrency for the Lambda functions.

B. Configure a higher provisioned concurrency for the Lambda functions

C. Convert the DB cluster to an Aurora global database Add additional Aurora Replicas in AWS Regions based on the locations of the company ' s customers.

D. Refactor the Lambda Functions Move the code blocks that initialize database connections into the function handlers.

E. Use Amazon RDS Proxy to create a proxy for the Aurora database Update the Lambda functions to use the proxy endpoints for database connections.

我的答案：_____

第 10 題 （題庫 No.370 · 單選）

A company provides an application to customers. The application has an Amazon API Gateway REST API that invokes an AWS Lambda function. On initialization, the Lambda function loads a large amount of data from an Amazon DynamoDB table. The data load process results in long cold- start times of 8-10 seconds. The DynamoDB table has DynamoDB Accelerator (DAX) configured. Customers report that the application intermittently takes a long time to respond to requests. The application receives thousands of requests throughout the day. In the middle of the day, the application experiences 10 times more requests than at any other time of the day. Near the end of the day, the application ' s request volume decreases to 10% of its normal total. A DevOps engineer needs to reduce the latency of the Lambda function at all times of the day. Which solution will meet these requirements?

A. Configure provisioned concurrency on the Lambda function with a concurrency value of 1. Delete the DAX cluster for the DynamoDB table.

B. Configure reserved concurrency on the Lambda function with a concurrency value of 0.

C. Configure provisioned concurrency on the Lambda function. Configure AWS Application Auto Scaling on the Lambda function with provisioned concurrency values set to a minimum of 1 and a maximum of 100.

D. Configure reserved concurrency on the Lambda function. Configure AWS Application Auto Scaling on the API Gateway API with a reserved concurrency maximum value of 100.

我的答案：_____

解答與解析　Answer Key

解析取自題庫原文，未經 AWS 官方審核；若與官方文件有出入，以官方文件為準。

第 1 題（No.6）　正確答案：C

To meet the requirements, the DevOps engineer should do the following:
Turn on the Multi-AZ option on the Aurora cluster.
Update the application to use the Aurora cluster endpoint for write operations.
Update the Aurora cluster ' s reader endpoint for reads.
Turning on the Multi-AZ option will create a replica of the database in a different Availability Zone. This will ensure that the database remains available even if one of the Availability Zones is unavailable.
Updating the application to use the Aurora cluster endpoint for write operations will ensure that all writes are sent to both the primary and replica databases. This will ensure that the data is always consistent.
Updating the Aurora cluster ' s reader endpoint for reads will allow the application to read data from the replica database. This will improve the performance of the application during the maintenance window.

涉及服務：RDS/Aurora

第 2 題（No.42）　正確答案：A

The correct solution is to create a new assertion safety rule in Route 53 ARC and apply it to the two routing controls. An assertion safety rule is a type of safety rule that ensures that a minimum number of routing controls are always enabled. The ATLEAST type of assertion safety rule specifies the minimum number of routing controls that must be enabled for the rule to evaluate as healthy. By setting the threshold to 1, the rule ensures that at least one routing control is always turned on. This prevents the scenario where both routing controls are accidentally turned off and the application becomes unavailable in both Regions.
The other solutions are incorrect because they do not use safety rules to prevent both routing controls from being turned off. A gating safety rule is a type of safety rule that prevents routing control state changes that violate the rule logic. The OR type of gating safety rule specifies that one or more routing controls must be enabled for the rule to evaluate as healthy. However, this rule does not prevent a user from turning off both routing controls manually. A resource set is a collection of resources that are tested for readiness by Route 53 ARC. A readiness check is a test that verifies that all the resources in a resource set are operational. However, these concepts are not related to routing control states or safety rules. Therefore, creating a new resource set and a new readiness check will not ensure that at least one routing control is turned on at all times. References:
Routing control in Amazon Route 53 Application Recovery Controller
Viewing and updating routing control states in Route 53 ARC
Creating a control panel in Route 53 ARC
Creating safety rules in Route 53 ARC

涉及服務：Route 53、ELB/ALB/NLB

第 3 題（No.54）　正確答案：A、B、C

* S3 RTC ensures #15-min replication SLA.
* Multi-Region Access Point (MRAP) simplifies active-passive replication access.
* SubmitMultiRegionAccessPointRoutes API manages routing changes during failover. This configuration follows AWS DR best practices for S3 active/passive architectures.

涉及服務：S3、Route 53

第 4 題（No.130） 正確答案：**D**

（此題題庫未附解析）

涉及服務：EC2/Auto Scaling、S3、Route 53、ELB/ALB/NLB

第 5 題（No.184） 正確答案：**A、C、E**

Short Explanation: To improve the performance of the application, the DevOps engineer should use Amazon RDS Proxy, implement the database connection opening outside the Lambda event handler code, and connect to the proxy endpoint from the Lambda function.

Amazon RDS Proxy is a fully managed, highly available database proxy for Amazon Relational Database Service (RDS) that makes applications more scalable, more resilient to database failures, and more secure1.

By using Amazon RDS Proxy, the DevOps engineer can reduce the overhead of opening and closing connections to the database, which can improve latency and throughput2.

The DevOps engineer should connect the proxy to the Aurora cluster reader endpoint, which allows read-only connections to one of the Aurora Replicas in the DB cluster3. This can help balance the load across multiple read replicas and improve performance for read-intensive workloads4.

The DevOps engineer should implement the database connection opening outside the Lambda event handler code, which means using a global variable to store the database connection object5. This can enable connection reuse across multiple invocations of the Lambda function, which can reduce latency and improve performance.

The DevOps engineer should connect to the proxy endpoint from the Lambda function, which is a unique URL that represents the proxy. This can allow the Lambda function to access the database through the proxy, which can provide benefits such as connection pooling, load balancing, failover handling, and enhanced security.

The other options are incorrect because:

Implementing database connection pooling inside the Lambda code is unnecessary and redundant when using Amazon RDS Proxy, which already provides connection pooling as a service.

Implementing the database connection opening and closing inside the Lambda event handler code is inefficient and costly, as it can increase latency and consume more resources for each invocation of the Lambda function.

Connecting to the Aurora cluster endpoint from the Lambda function is not optimal for read-only queries, as it can direct traffic to either the primary instance or one of the Aurora Replicas in the DB cluster. This can result in inconsistent performance and potential conflicts with write operations on the primary instance.

涉及服務：Lambda、RDS/Aurora、EventBridge、ELB/ALB/NLB

第 6 題（No.212） 正確答案：**D**

EventBridge is needed to detect the database failure. Lambda is needed to promote the replica as it ' s in another Region (manual promotion, otherwise). Storing and updating the endpoint in Parameter store is important in updating the application. Look at High Availability section of Aurora FAQ: https://aws.amazon.

com/rds/aurora/faqs/

涉及服務：CloudFormation、Systems Manager (SSM)、Lambda、CloudWatch、CloudTrail、RDS/Aurora

第 7 題（No.254） 正確答案：**A**

ALB natively supports WebSocket protocols and sticky sessions via target group session affinity.

Deploying ALBs in multiple regions with cross-zone load balancing ensures high availability and fault tolerance.

Using Route 53 latency-based routing allows users worldwide to connect to the lowest latency region, minimizing delay.

NLBs do not support sticky sessions and WebSocket protocol as well as ALBs do. Combining ALBs and

NLBs (Option D) increases complexity. CloudFront (Option C) does not natively support WebSocket sticky sessions.
Option A is the simplest, most effective solution meeting all requirements with least operational overhead.
References:
ALB WebSocket Support
Route 53 Latency Based Routing
涉及服務：Route 53、ELB/ALB/NLB

第 8 題（No.296） 正確答案：B

https://docs.aws.amazon.com/autoscaling/ec2/userguide/ASGettingNotifications.html#auto-scaling-sns-notifications
涉及服務：Lambda、EC2/Auto Scaling、CloudWatch、SNS

第 9 題（No.302） 正確答案：B、E

The correct answer is B and E. Configuring a higher provisioned concurrency for the Lambda functions will ensure that the functions are ready to respond to the initial burst of requests without any cold start latency.
Using Amazon RDS Proxy to create a proxy for the Aurora database will enable the Lambda functions to reuse existing database connections and reduce the overhead of establishing new ones. This will also improve the scalability and availability of the database by managing the connection pool size and handling failovers.
Option A is incorrect because reserved concurrency only limits the number of concurrent executions for a function, not pre-warms them. Option C is incorrect because converting the DB cluster to an Aurora global database will not address the issue of database connection latency, and may introduce additional costs and complexity. Option D is incorrect because moving the code blocks that initialize database connections into the function handlers will not improve the performance or scalability of the Lambda functions, and may actually worsen the cold start latency. References:
AWS Lambda Provisioned Concurrency
Using Amazon RDS Proxy with AWS Lambda
Certified DevOps Engineer - Professional (DOP-C02) Study Guide (page 173)
涉及服務：Lambda、S3、RDS/Aurora、API Gateway

第 10 題（No.370） 正確答案：C

The following are the steps that the DevOps engineer should take to reduce the latency of the Lambda function at all times of the day:
Configure provisioned concurrency on the Lambda function.
Configure AWS Application Auto Scaling on the Lambda function with provisioned concurrency values set to a minimum of 1 and a maximum of 100.
The provisioned concurrency setting ensures that there is always a minimum number of Lambda function instances available to handle requests. The Application Auto Scaling setting will automatically scale the number of Lambda function instances up or down based on the demand for the application.
This solution will ensure that the Lambda function is able to handle the increased load during the middle of the day, while also keeping the cold-start latency low.
The following are the reasons why the other options are not correct:
Option A is incorrect because it will not reduce the cold-start latency of the Lambda function.
Option B is incorrect because it will not scale the number of Lambda function instances up or down based on demand.
Option D is incorrect because it will only configure reserved concurrency on the API Gateway API, which will not affect the Lambda function.

涉及服務：Lambda、EC2/Auto Scaling、DynamoDB、API Gateway